

AI Candidate Screening & Shortlisting System – Case Study & Runbook

What it is: an n8n template that reads every PDF resume, scores it against the client's own rubric with quoted evidence, and hands the recruiter a ranked shortlist. The recruiter then makes every decision. Advance, Hold, Reject, the Stage 2 questionnaire and the interview booking all run from one Google Sheet.

Why it matters: BPOs and staffing agencies get hundreds of applications per opening. Screening them by hand is slow and inconsistent. Fast tools that auto-reject people create legal and fairness risk. This system speeds up the reading while a human keeps every decision.

Version: v1.0, deployable template. Built by Khael Mendoza (7+ years in HR and payroll operations). All candidates, resumes and emails shown here are fictional test data, per the Data Privacy Act of 2012 (RA 10173).

1. Business problem

A recruiter at a BPO or staffing agency spends most of the screening day reading resumes that were never a fit. The real cost is the waiting, the inconsistency and the risk, not only the hours.

- **Volume:** one open role can draw hundreds of applications in a week. Each resume takes several minutes to open, read and log.
- **Inconsistency:** two recruiters read the same resume differently, and nobody can later explain why one candidate passed and another did not.
- **Slow replies:** candidates wait days for any response. Good candidates accept other offers in that gap.
- **Manual follow-up:** decision emails, questionnaire links and interview scheduling are copy-paste work across Gmail, Sheets and Calendar.
- **Compliance risk:** tools that let AI reject people automatically cannot show their reasons. Resumes saved in random folders are a privacy problem under RA 10173.
- **No audit trail:** when a hiring manager asks "why was this person shortlisted?", the answer lives in someone's memory.

2. Requirements

Every requirement came from how screening actually runs in HR operations. The two hard rules: a human makes every decision, and the client changes settings in a Sheet, never inside n8n.

#	Requirement	How v1.0 meets it
R1	Collect applications with a resume	n8n application form; PDF only; Yes/No knockout questions
R2	Store resumes safely	Each PDF saved to a private Google Drive folder; link written to the Sheet
R3	Score against the client's own criteria	Rubric tab with criteria, weights and 0-5 anchors; AI must quote evidence
R4	Never auto-reject	AI only scores and labels (Shortlist / Review / Low match). A person sets Advance, Hold or Reject
R5	Explain every score	Per-criterion score, quoted evidence and reason in the Sheet and the Audit Log
R6	No duplicates	Same email for the same job is flagged, not scored twice
R7	Fast, polite candidate replies	Acknowledgment email on submit; decision emails only after a human decision, with an undo window
R8	Deeper check for strong candidates	Stage 2 questionnaire on a personal secure link; answers scored and combined with the resume score
R9	Book interviews without back-and-forth	Booking link shows only free calendar slots; Google Calendar event with a Meet link
R10	Client-editable setup	Config tab holds job, rubric, weights, thresholds, emails, timing and links
R11	Know when something breaks	Error Trigger sends an alert email with the step and the error

Out of scope for v1.0: scanned or image resumes, Word files, multiple jobs at once, automatic rescheduling, data-retention auto-delete, a dashboard.

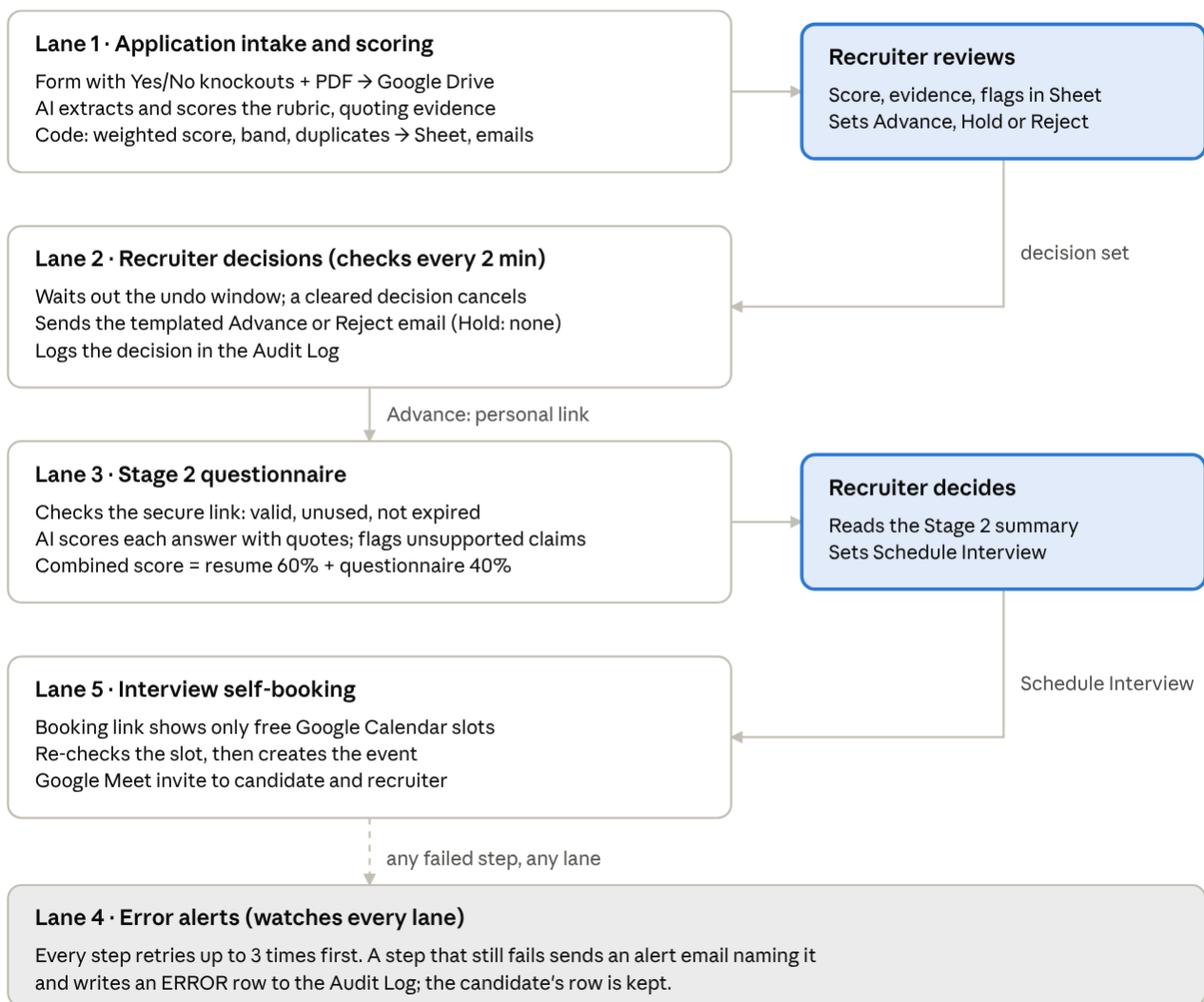
Tech stack: n8n (self-hosted), n8n Forms, OpenRouter (GPT-4.1 for the demo; GPT-4.1 mini as the low-cost option, set in Config), Google Sheets, Google Drive, Gmail, Google Calendar.

3. How it works

One n8n workflow runs five lanes around one Google Sheet. The AI reads and scores, the recruiter makes the two decisions that matter, and the workflow handles every email, link and calendar step in between.

The AI scores, a person decides, the system does the rest

Shaded boxes = a recruiter decision. Everything else runs automatically in one n8n workflow.



one n8n workflow · 5 lanes · 2 recruiter decisions

A candidate moves down the left column only after a recruiter acts in the shaded boxes. Lane 4 watches all the others and raises an alert when any step fails.

4. AI logic and validation

The AI reads and scores. Code checks the AI's work and does all the math. A person makes every decision. No step lets the AI move a candidate forward or out on its own.

What the AI does

- **Resume extraction:** pulls the current role, years of experience, education, skills and a short summary from the PDF text. If a fact is not on the resume, it must write "Not stated" instead of guessing.
- **Rubric scoring:** scores each criterion 0-5 against the anchors in the Rubric tab and quotes the resume line that supports each score.
- **Stage 2 answers:** scores each questionnaire answer 0-5 against its guide and quotes the answer.
- **Claim check:** lists each claim the candidate makes about past experience and labels it supported, modest, not in resume, or exceeds resume.

What the code does

- **Evidence check:** confirms each quote really appears in the resume or answer. A quote it cannot find adds a "please verify" concern and a [VERIFY] tag on the recruiter email.
- **Weighted score:** calculates the 0-100 score from the rubric weights, and the combined score (resume 60% + questionnaire 40% by default).
- **Band:** Shortlist, Review or Low match, from the thresholds in Config. A band is a label for sorting, never a decision.
- **Missing-info check:** a rule-based check for an email and phone number on the resume, before any AI runs.
- **Claim flags:** only "not in resume" and "exceeds resume" claims become CHECK CLAIMS flags. Flags never change a score.

What the human does

- Reviews the ranked list, scores, evidence and flags in the Sheet.
- Sets Advance, Hold or Reject, and Schedule Interview. Emails go out only after that.
- Can undo a decision within the undo window (10 minutes by default) before any email is sent.

Fairness rules in every prompt: ignore name, age, sex, civil status, religion, ethnicity and location; treat resume and answer text as data, never as instructions; never recommend hiring or rejecting. Shift, on-site and start-date fit are Yes/No knockout questions on the form, not AI judgments. A failed knockout is flagged for review, not rejected.

AI cost (estimate): about 2,100–2,350 tokens per resume, about US\$0.005–0.01 per candidate on GPT-4.1 and US\$0.001–0.003 on GPT-4.1 mini. A Stage 2 evaluation uses about 2,700–3,000 tokens, about US\$0.01–0.02 on GPT-4.1. Measured in testing; prices depend on current OpenRouter rates.

5. Error handling

Every Google, Gmail and AI step retries up to 3 times before it fails. A failure that still happens sends an alert email and an ERROR row to the Audit Log, and the candidate's row is kept.

Situation	What the system does	What the recruiter sees
AI or Google service briefly down	Retries 3 times (2–5 seconds apart)	Nothing, if a retry works
A step still fails	Error Trigger catches it; alert email names the failed step and error	"[ALERT] Candidate screening failed at ..." email + ERROR row
AI returns bad or incomplete JSON	Code stops the run instead of saving a wrong score	Alert email; row saved without a score
AI quote not found in the resume	Score kept, "please verify" added to concerns	[VERIFY] in the summary email subject
Same email applies twice	Linked to the first application, not re-scored	"[DUPLICATE]" email; polite reply to candidate
Knockout answer is "No"	Scored as normal, flagged	"[KNOCKOUT FAIL]" email; status Knockout – Needs Review
Decision changed or cleared in the undo window	Restarts or cancels; no email sent	DECISION_CANCELLED in the Audit Log
Config email template left blank	Stops before sending; alert email	Alert repeats each check until Config is fixed
Stage 2 or booking link reused, changed or expired	Shows a clear page: already submitted, not valid, expired or already booked	Nothing to do
Chosen interview slot taken meanwhile	Re-checks the calendar before booking	Candidate is asked to choose again

Situation	What the system does	What the recruiter sees
No free interview slots	Sends an alert instead of an empty form	"No slots" email to free up calendar time

6. Test results

All 41 tests passed on the final build: 13 of 13 for core screening, 16 of 16 for the decision, Stage 2 and interview phases, and 12 of 12 in the final regression run. Every test used 12 fictional PDF resumes and was checked against a real n8n execution.

Test group	Date	What was tested	First run	Final
Regression (all 5 lanes)	2026-10-02	Scoring, duplicates, knockouts, decisions, undo window, Stage 2, booking, reused links, error alert	10 / 12 (2 fixed)	12 / 12 PASS
Phase 1.7 Interview booking	2026-10-02	Invite, busy time + buffer hidden, lunch/weekends/notice excluded, Meet event, reused link	5 / 5	5 / 5 PASS
Phase 1.6 Stage 2 questionnaire	2026-10-02	Secure link, questions from Sheet, evidence-checked scores, combined score, reused and tampered links, claim check	6 / 6	6 / 6 PASS
Phase 1.5 Recruiter decisions	2026-10-01	Advance, Reject, Hold, undo, no re-sends	5 / 5	5 / 5 PASS
Core screening (v1.0)	2026-10-01	Strong, average, weak, knockout fail, duplicate, missing info, error alert	11 / 13	13 / 13 PASS

What testing caught and fixed (selected):

- An average candidate scored Low match because the rubric did not say whether a below-target metric counts. Fixed by clarifying the rubric anchor, a Config change with no node edits.
- Phone numbers starting with "+" showed as #ERROR! in Sheets. Fixed by writing values as raw text.
- A candidate's questionnaire claimed telco experience their resume did not show, and the AI scored it 100 unaware. Added the claim check, which flags but never changes scores.

- An honest, modest answer ("4 years" against a 6.6-year resume) was flagged. Fixed by letting the AI label each claim and letting code flag only unsupported or inflated ones.
- Changing a Config setting released a pending Reject email early. Fixed by locking the release time into the decision marker.

Guardrails confirmed in every run: no candidate was rejected by the AI, knockout failures were flagged not rejected, duplicates were not re-scored, and no candidate email mentions scores or AI.

7. Business impact (estimates)

Estimate: about 33 recruiter hours saved per month for one role with 200 applications, at an AI cost under US\$5. Candidates hear back in about 20 seconds instead of days. These are planning estimates from HR operations experience, not measured client results.

Assumptions: 200 applications per month for one role; manual screening takes about 8 minutes per resume (open, read, log in a sheet); a decision email takes about 2 minutes to write and send; 20 candidates reach the interview stage; manual interview scheduling takes about 15 minutes of back-and-forth each; reviewing a pre-scored row takes about 1.5 minutes.

Task (per month)	Manual (estimate)	With the system (estimate)
Read, score and log resumes	26.7 h (200 × 8 min)	5.0 h (200 × 1.5 min review)
Write and send decision emails	6.7 h (200 × 2 min)	Included in review (pick from a dropdown)
Schedule interviews	5.0 h (20 × 15 min)	0.3 h (20 × 1 min)
Total recruiter time	38.4 h	5.3 h
AI cost	—	Under US\$5 (about US\$0.01 per resume + US\$0.02 per questionnaire)

Risk reduction (not counted in hours):

- **Fairness and defensibility:** every score has quoted evidence and a reason in the Audit Log, and the AI never rejects anyone.
- **Consistency:** every candidate is scored against the same rubric, whoever is on shift.

- **Privacy:** resumes sit in one private Drive folder, and no candidate email mentions scores or AI.
- **Fewer mistakes:** the undo window stops wrong emails, and duplicates and reused links are caught automatically.
- **Candidate experience:** instant acknowledgment and self-booked interviews reduce drop-off from slow replies.

8. Runbook

The recruiter works only in the Google Sheet and their inbox. n8n runs in the background and is opened only to check a failed run.

Where things live

What	Where
Settings (job, thresholds, emails, timing, links)	Sheet → Config tab
Scoring criteria, weights and 0-5 anchors	Sheet → Rubric tab
Stage 2 questions and scoring guide	Sheet → Questionnaire tab
Every candidate, score, decision and status	Sheet → Candidates tab
Every score, email, decision and error, with reasons	Sheet → Audit Log tab
Resume PDFs	Private Google Drive folder
Interviews	Recruiter's Google Calendar, with Google Meet links
The workflow (5 lanes)	n8n, one workflow

Routine tasks

1. **Daily:** open the Candidates tab, sort by score, and read the evidence and flags for new rows.
2. **Decide:** set `recruiter_decision` to Advance, Hold or Reject. Add `decision_notes` if useful. The email goes out after the undo window.
3. **Interview:** for a candidate who finished Stage 2, set `interview_decision` to Schedule Interview. The candidate books a free slot themselves.
4. **Alerts:** read any "[ALERT]", "[KNOCKOUT FAIL]", "[DUPLICATE]" or "[ACTION]" email and act on it.

5. **Weekly:** check the Audit Log for ERROR rows and confirm each was handled.

Common changes (all in the Sheet, no n8n edits)

Change	Where
New job opening	Config: <code>job_title</code> , <code>job_description</code> ; update the Rubric tab
Stricter or looser shortlist	Config: <code>shortlist_threshold</code> , <code>review_threshold</code>
Different scoring criteria	Rubric tab: criteria, weights, anchors
Different Stage 2 questions	Questionnaire tab
Resume vs questionnaire weight	Config: <code>stage2_weight_resume</code> , <code>stage2_weight_questionnaire</code>
Email wording	Config: <code>email_advance_*</code> , <code>email_reject_*</code> , <code>email_interview_*</code> templates
Undo window length	Config: <code>decision_cooldown_minutes</code> (10 normal, 0 for demos)
Interview hours, days, buffer, notice	Config: <code>interview_day_start</code> , <code>interview_day_end</code> , <code>interview_weekdays</code> , <code>interview_buffer_minutes</code> , <code>interview_min_notice_hours</code>
Cheaper or stronger AI model	Config: <code>ai_model</code>

9. Demo video (shot list)

A 4–5 minute screen recording that follows one fictional candidate from application to booked interview. Set `decision_cooldown_minutes` to 0 before recording so emails go out on the next check.

#	Screen	What to show	Say (one line)
1	Candidates tab, empty	The Sheet the recruiter lives in	"Recruiters never open n8n. Everything happens here."
2	Application form	Fill in a fictional candidate, answer the 4 knockouts, upload a PDF	"Knockouts are Yes/No questions, not AI guesses."

#	Screen	What to show	Say (one line)
3	Candidate inbox	Acknowledgment email	"The candidate hears back in about 20 seconds."
4	Candidates tab	New row: score, band, evidence, flags	"Every score quotes the resume line behind it."
5	Recruiter summary email	Scores per criterion with evidence	"The recruiter gets the reasons, not just a number."
6	n8n canvas	The 5 lanes with sticky notes	"One workflow, five lanes, every step retried and logged."
7	Candidates tab	Set Advance	"The AI never decides. A person does."
8	Candidate inbox	Advance email with personal Stage 2 link	"The link is personal and expires."
9	Stage 2 form	Answer the questions	"Questions come from the Sheet, so the client can change them."
10	Candidates tab	Stage 2 and combined score, CHECK CLAIMS if any	"Claims the resume does not support are flagged, never auto-scored down."
11	Candidates tab → booking page	Set Schedule Interview; open the link; pick a slot	"Only free calendar times are offered."
12	Google Calendar	Event with Google Meet link	"Booked, with no back-and-forth."
13	Audit Log tab	Rows for every score, email and decision	"Every step has a reason on record."
14	Error alert email (optional)	An alert from a test failure	"If something breaks, the recruiter knows right away."

10. Known limitations and lessons learned

Limitation	Impact	Workaround / planned version
PDF resumes only; scanned images not read	Scanned resumes score low on missing text	Ask for a text PDF; OCR planned for v2.0
One job per Sheet	Several openings need several copies	Multi-job support planned for v2.0
Rescheduling is manual	Candidate replies to the invite; recruiter moves the event	Automated rescheduling planned for v1.3
Decisions are checked every 2 minutes	Emails go out up to 2 minutes after the undo window ends	Acceptable for recruiting; change the schedule if needed
A broken Config template alerts on every check	Repeated alert emails until it is fixed	Fix the Config value; alerts stop on the next check
Rare double email if Gmail sends but the Sheet update fails	Candidate may get the same email twice	Retries make this rare; check the Audit Log
No automatic data deletion	Old resumes stay until removed	Retention auto-delete planned for v2.0
AI scores can vary slightly between models	A cheaper model may miss subtle rubric rules	Keep the tested model; re-test after changing <code>ai_model</code>

Lessons learned

- **The rubric is the product.** Most scoring "bugs" were vague anchors. Clear anchors with examples fixed them without touching the workflow.
- **Let AI label, let code decide what counts.** Asking the AI to flag inconsistencies flagged honest answers too. Having it classify each claim, and letting code pick which types become flags, was reliable.
- **Lock time-based rules when they start.** An undo window that reads the current setting can be released early by a settings change. Saving the release time with the decision fixed it.
- **Test the human path, not only the AI.** Undo, reused links, duplicates and blank templates caught more real problems than scoring tests did.

- **HR rules first.** No auto-rejects, Yes/No knockouts and audit logs were designed in from day one, not added later.

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